



Battleship is an online version of the classic battle game reimagined using rainy-city.com style graphics and animations. The isometric view brings to life a new version of battleship never before seen. The exciting turn based combat comes to life through particle destruction animations, interactive maps, and a player leaderboard challenging players to play time after time to see who rises to the top. Battleship is a classic game now living in your web browser. Play today!

frontend

Battleship 3d model asset with
cannons firing animation periodically

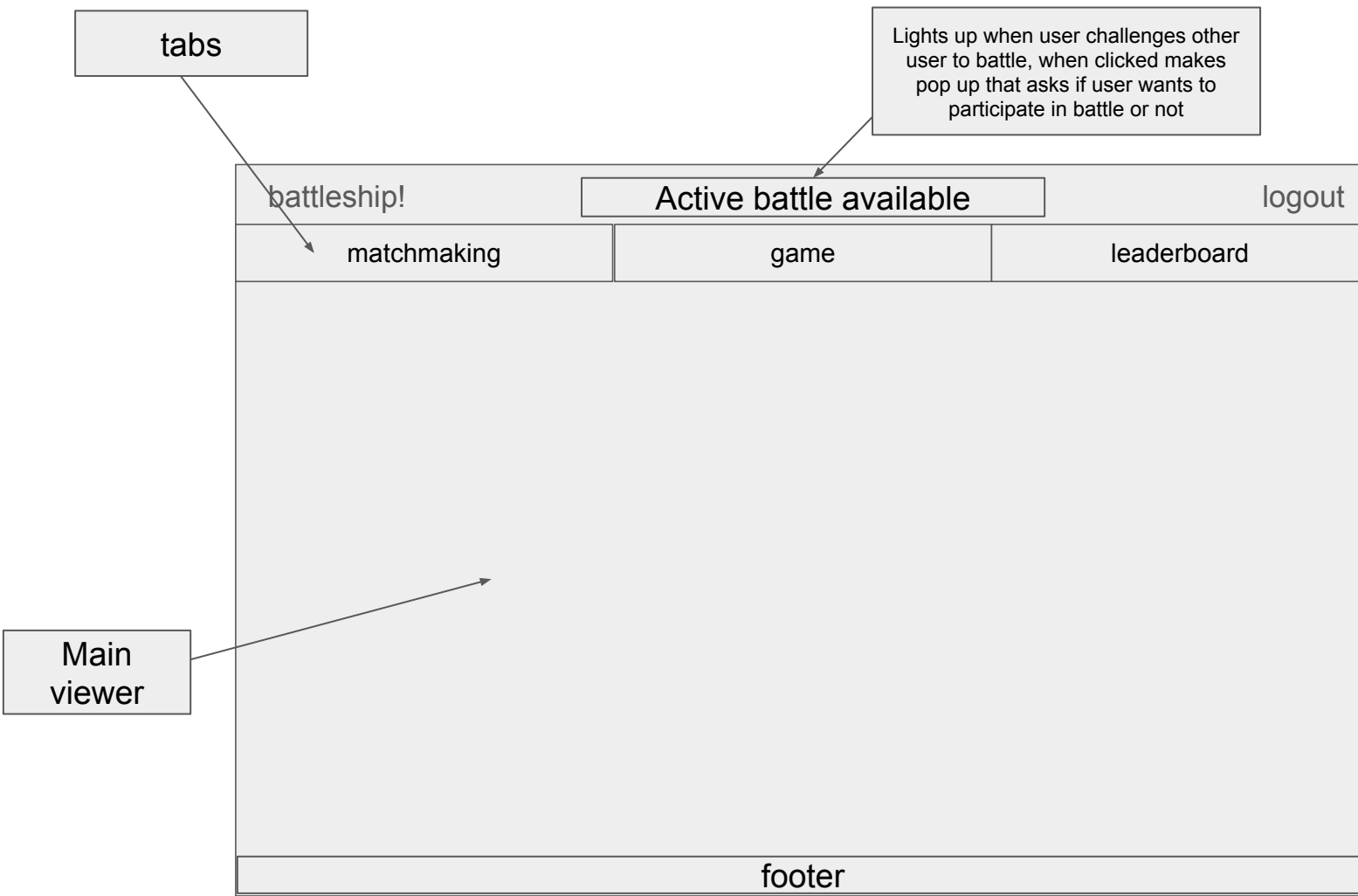
Battleship game

username

password

Create new user

Start page

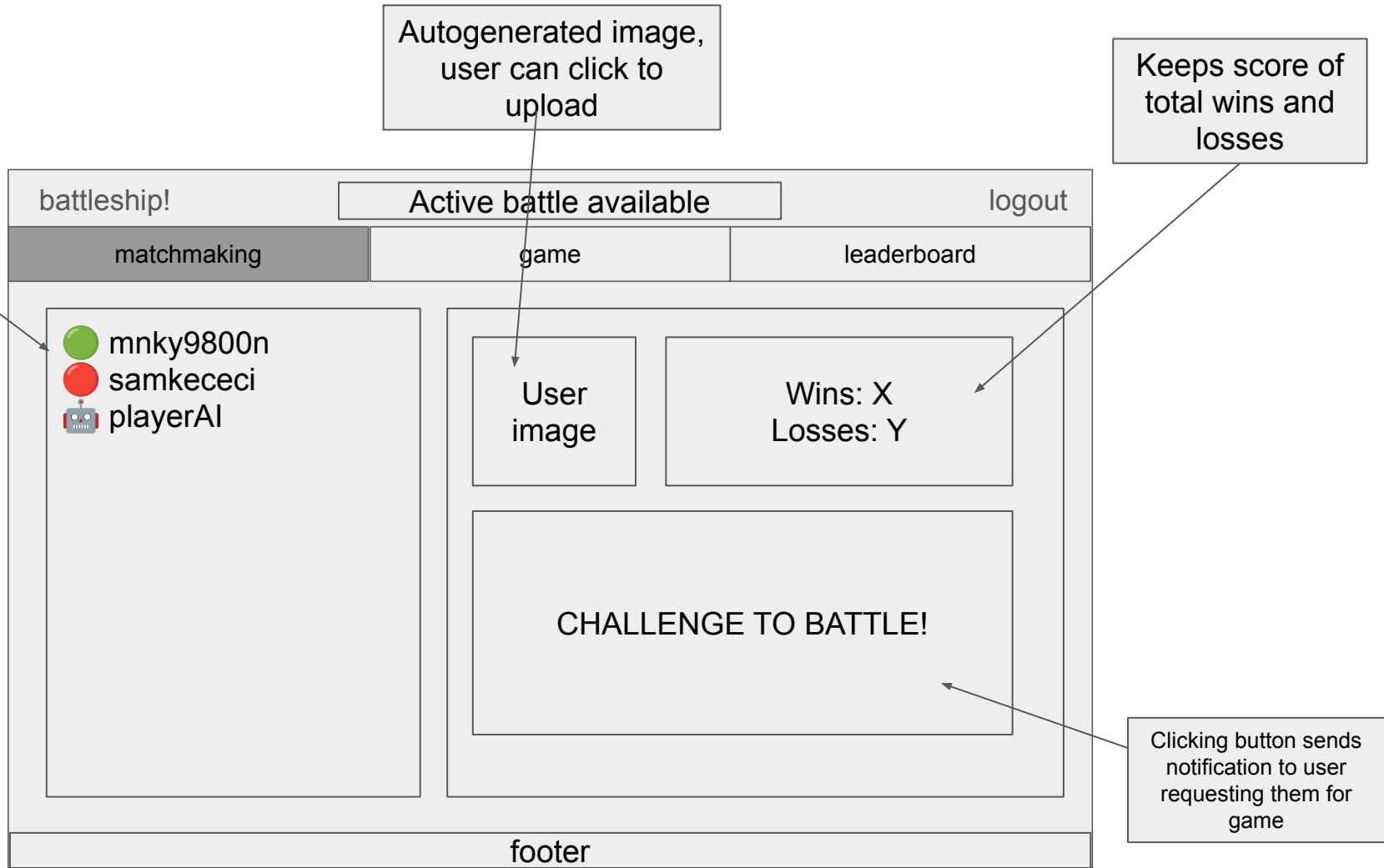


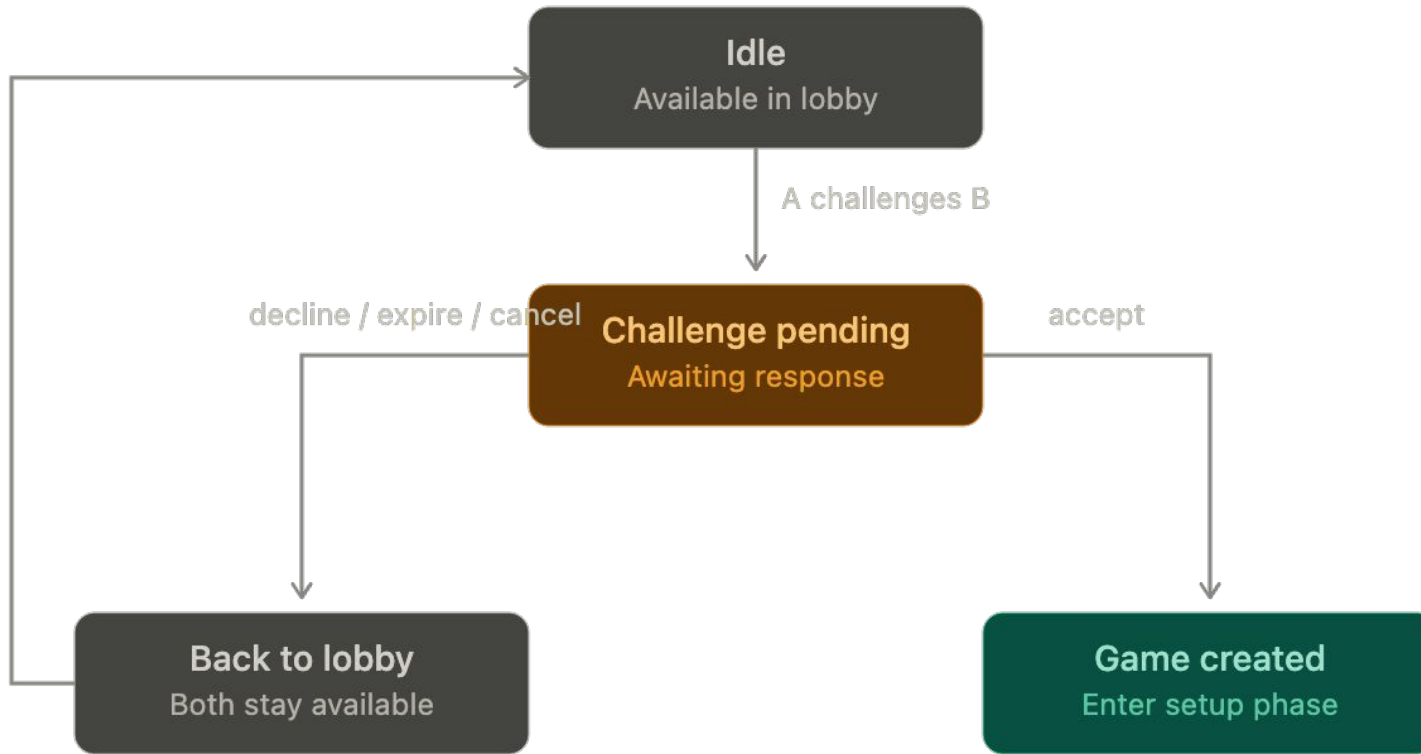
Login view, starts on matchmaker

● means user is online
● means user is not online
🤖 means user is AI and not real person

List has all users registered show up here, scrollable if long

Clicking on user name activates user viewport to right which gives more info about the user with that user name and allows the active player to challenge them to a battle



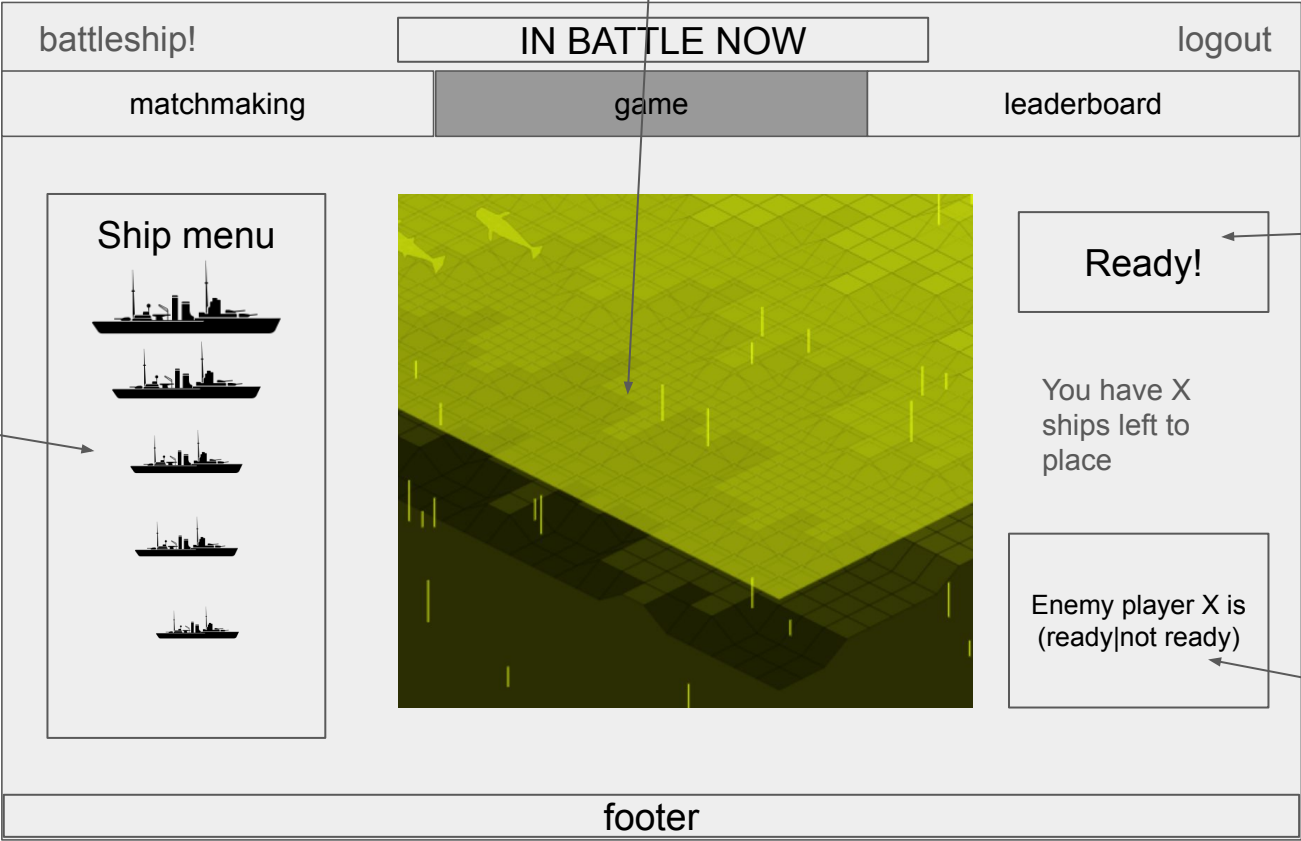


Decision tree for what happens when a user challenges another user to a game

Battle map
User can drag ships placed on map around the map, scroll wheel makes map zoom in and out, start view shows entire map in the window, whales are animated swimming around

Ready button
Button is greyed out unless all ships are on the map
When button is pressed other player is notified in the "enemy player is ready" message.

Ship menu
Uses actual art assets of game
Player can drag and drop on the map
Makes splash when ship is placed



Enemy readiness status
System checks if enemy player is ready based on server status
X is enemy username
Lights up and reports ready when enemy player has pressed the ready button

Gameview during setup

Active turn status

If enemy turn, left is lit up
If players turn, right is lit up

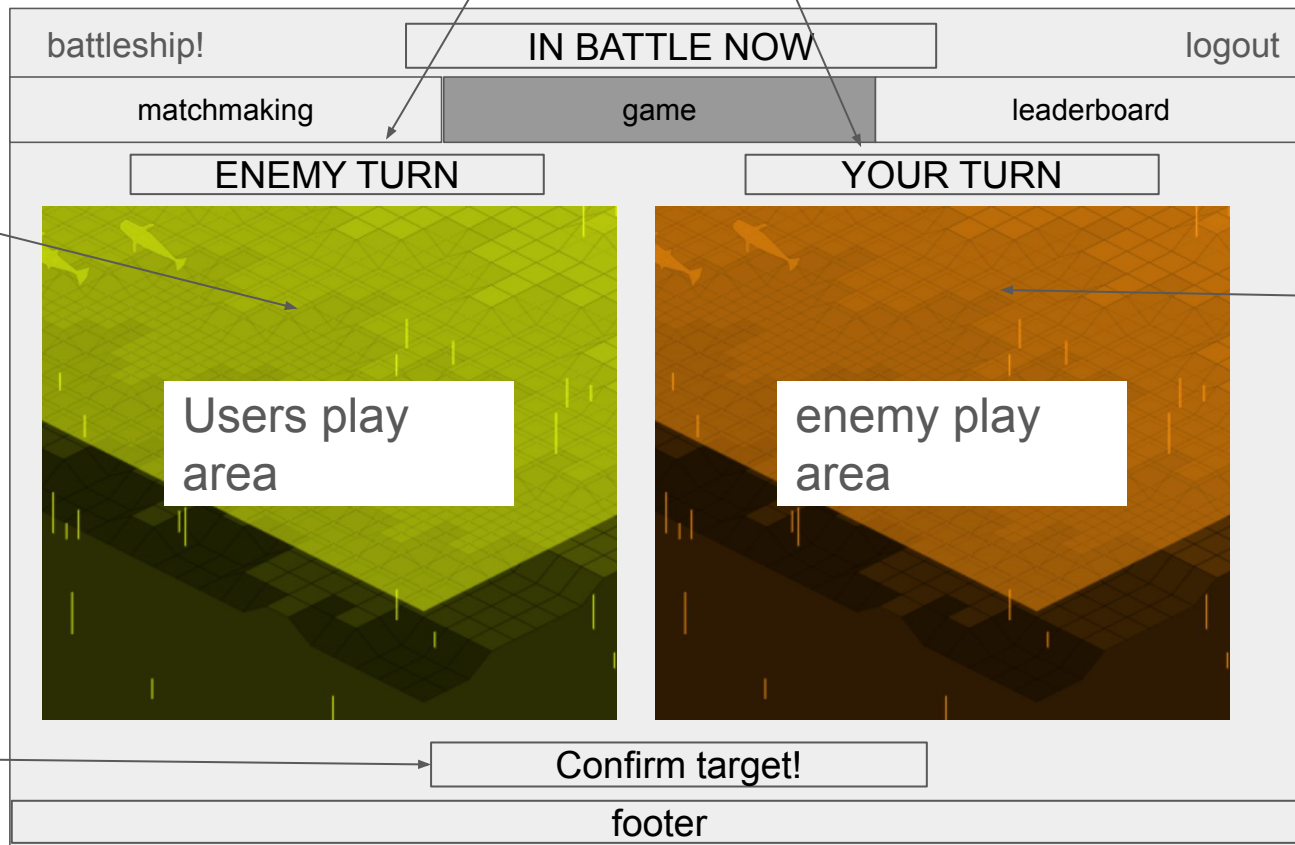
Interactions

During enemy
players turn this
map is active

Destruction
animation from
rainy-city.com
plays when
enemy selects
a target and
fires

Confirm target
button

Greyed out unless
players turn AND
player has selected
a grid square to
target
If grid square
selected then button
lights up



Interactions

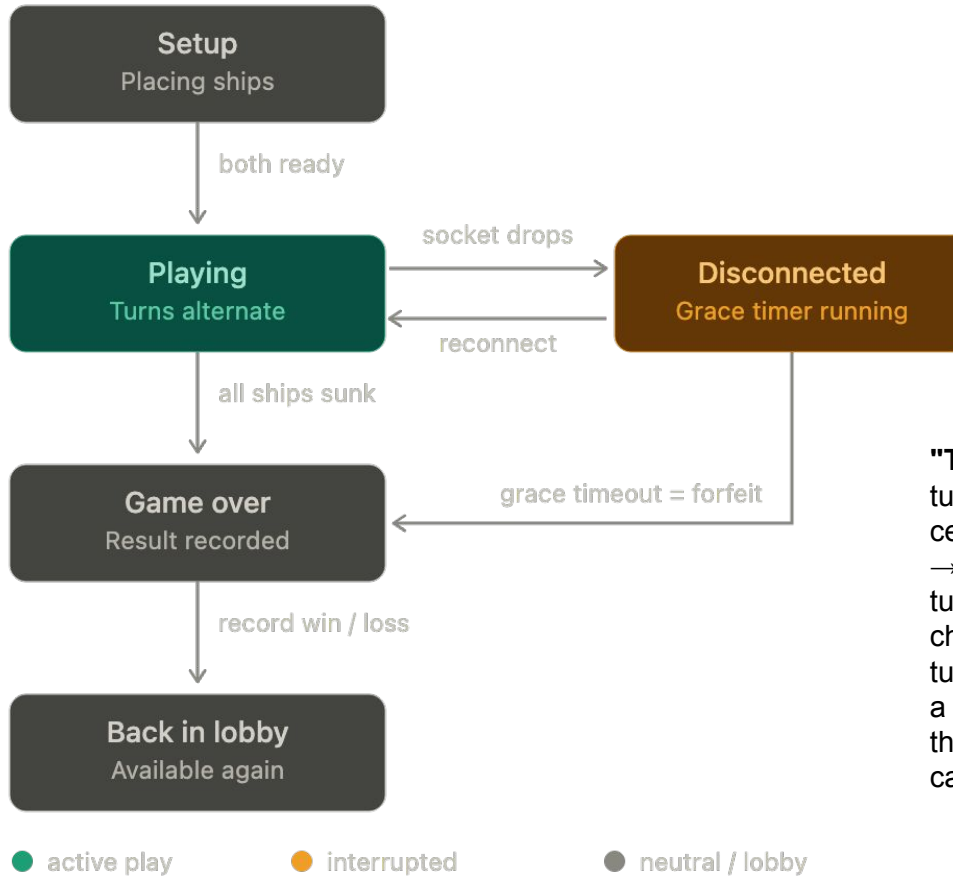
During players turn this
map is active

Destruction animation
from rainy-city.com plays
when player selects a
target and fires (presses
confirm target button)
IF target is successful,
destruction animation
plays 5 times, ELSE only
one time

IF ship is destroyed then
the ship is shown on the
map but recolored RED

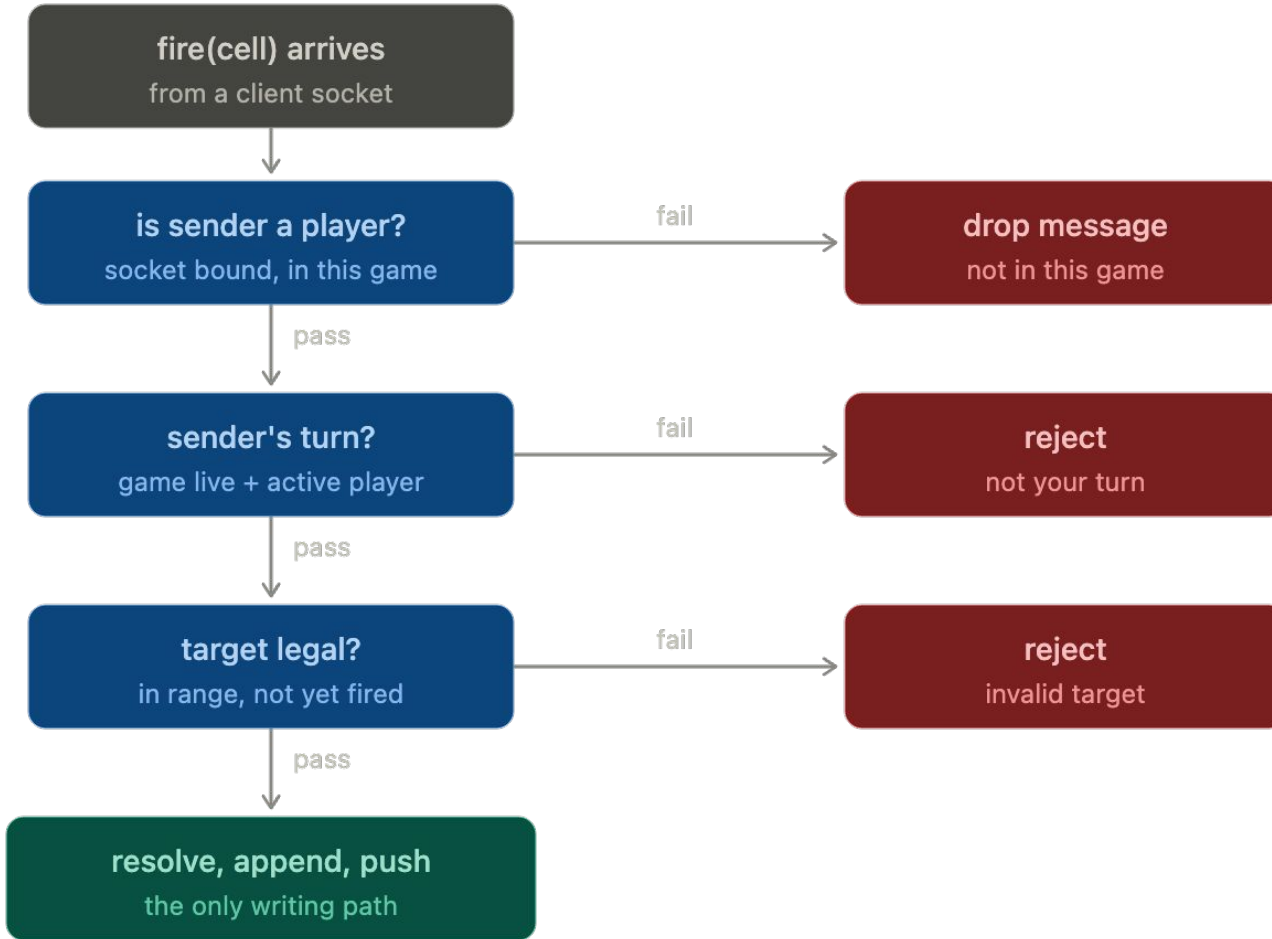
Notes: if the game is won (all of the ships of one player destroyed,
checked each turn) then a pop up happens that says YOU WON|LOST!
Click to continue, and players are taken back to the matchmaking screen.

Game view during gameplay



"Turns alternate" hides a self-loop. Each turn is: validate it's your turn → validate the cell wasn't already hit → resolve hit/miss/sink → check win → swap active player. Most turns loop back into "playing"; only the win check exits to "game over." There's also a turn timer living inside this state that triggers a 15 second countdown after 45 seconds on the grace timer — it's the other way a game can end that isn't "all ships sunk."

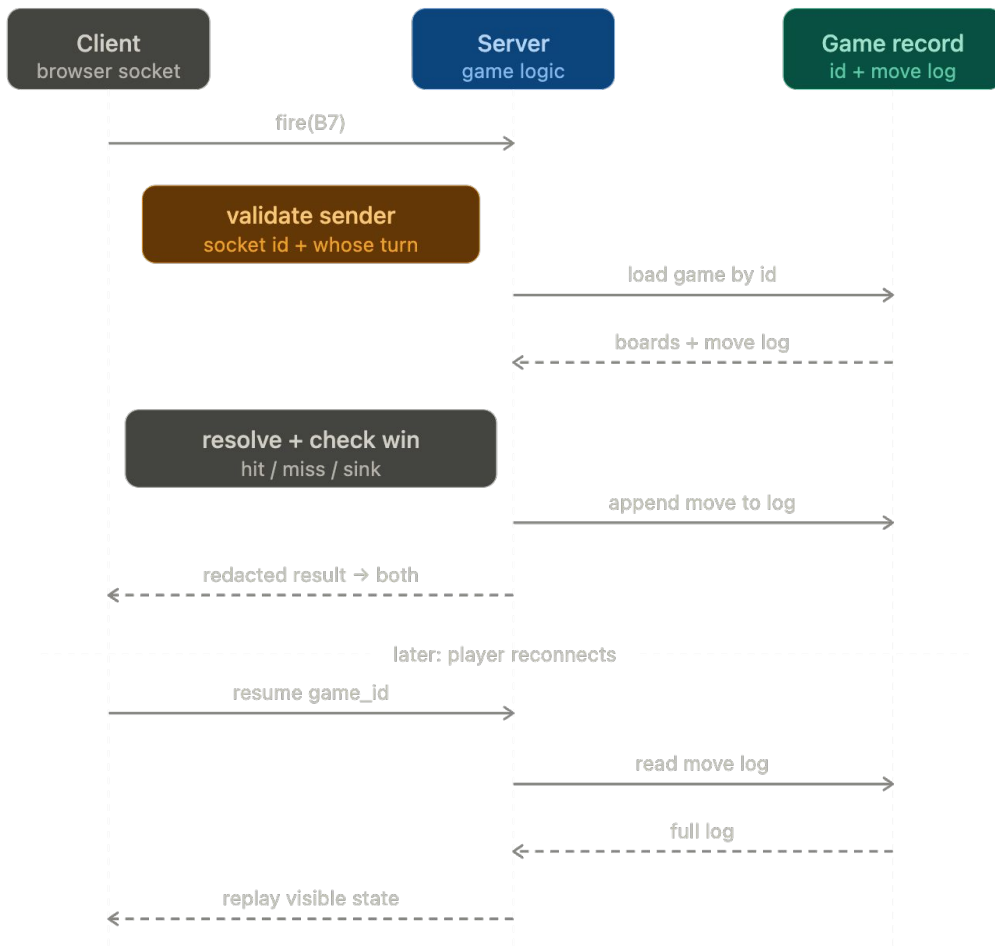
Decision tree for in-game



● check

● rejected

● writes to the log



Authenticating and
recording who fired

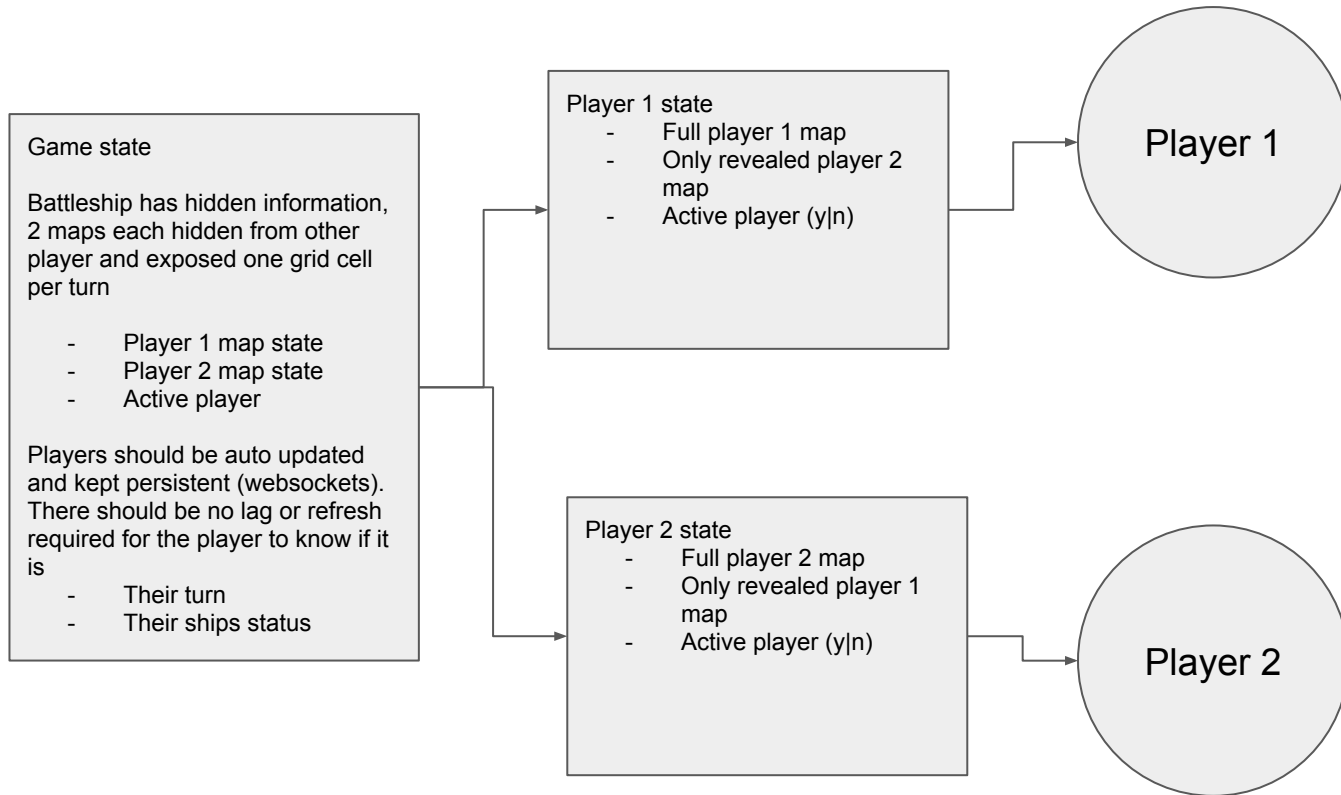
battleship!		Active battle available		logout
matchmaking		game		leaderboard
PLAYER  mnky9800n  samkececi  playerAI		WINS 30 20 10		LOSSES 10 10 20
footer				

Ordered by max
wins

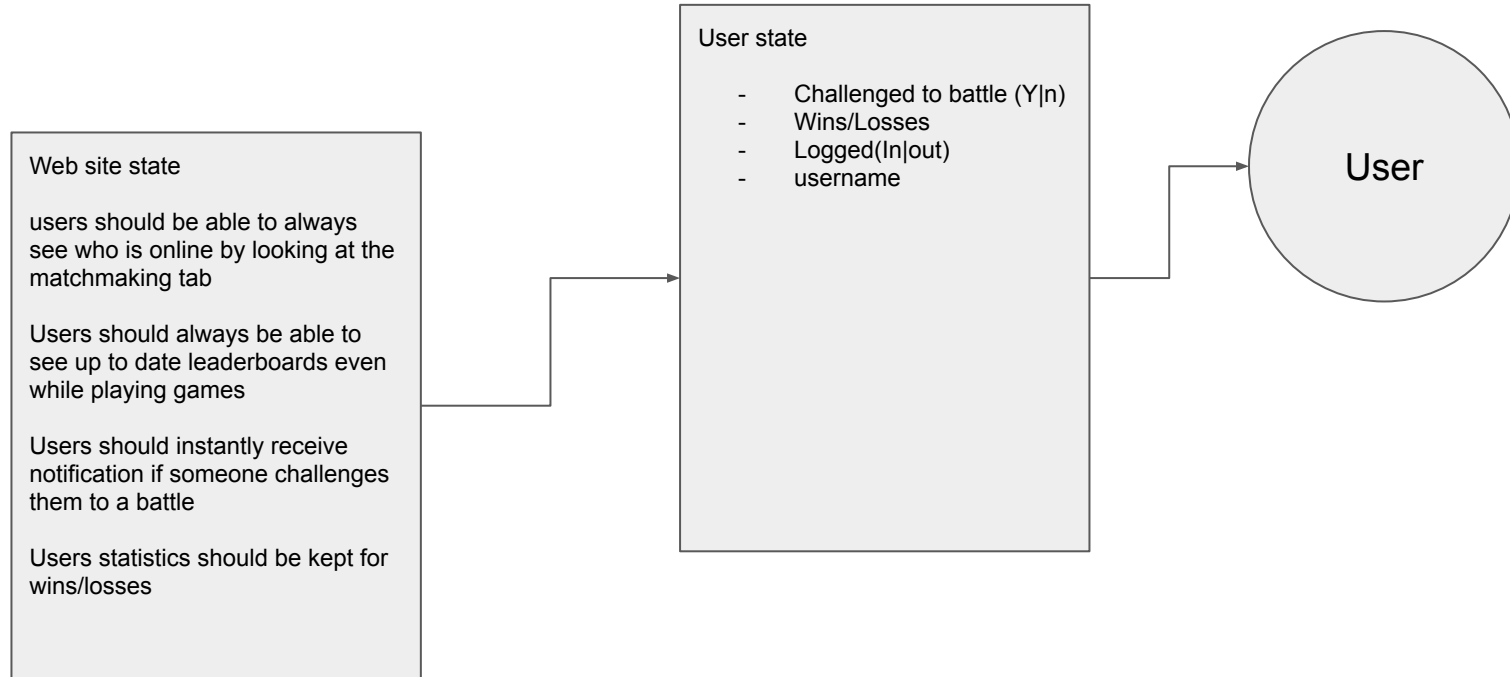
LEADERBOARD VIEW

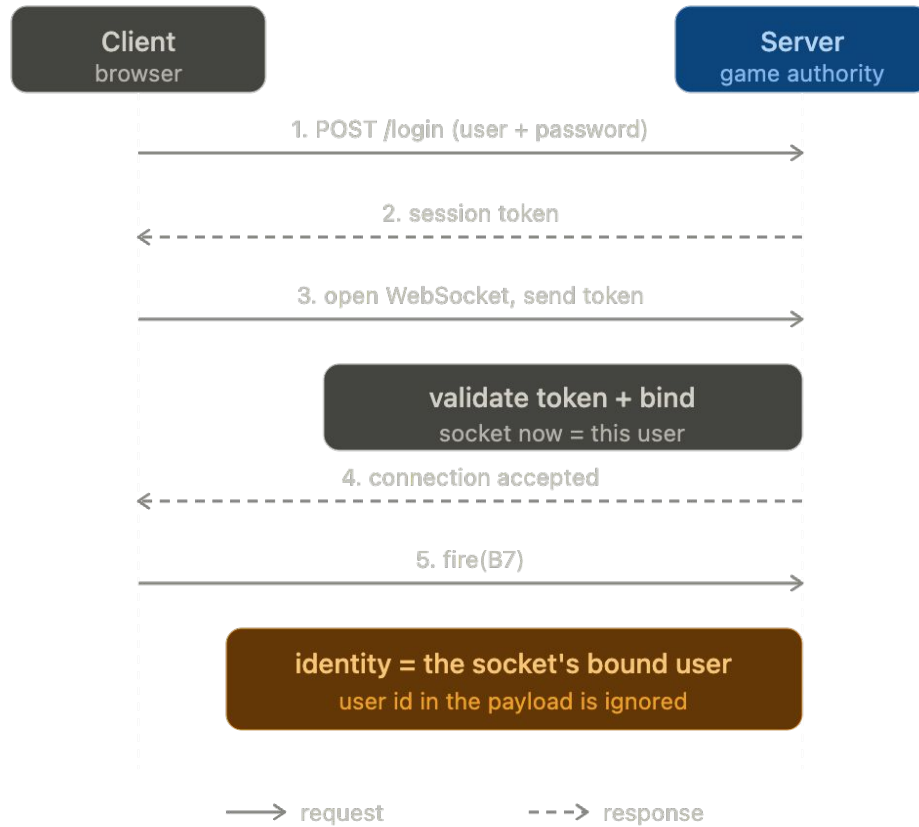
backend/states/process

In game information



Out of game information





Authenticating a user

CLASSIC BATTLESHIP BOARD GAME RULES ONLY FOR REFERENCE

Setup

Each player has two 10×10 grids, with columns labeled A–J and rows numbered 1–10. One grid is for placing your own fleet; the other is for tracking your shots at the enemy. The grids are hidden from your opponent.

Each player has a fleet of 5 ships, placed secretly on their own grid:

- Carrier — 5 cells
- Battleship — 4 cells
- Cruiser — 3 cells
- Submarine — 3 cells
- Destroyer — 2 cells

Ships are placed horizontally or vertically (never diagonally), must lie entirely on the grid, and may not overlap each other. In the classic rules ships *may* sit directly adjacent to one another.

Playing a turn

Players alternate turns. On your turn you call out a single coordinate (e.g. "B7") as a shot at the opponent's fleet. Your opponent announces whether it's a **hit** or a **miss**.

You mark the result on your tracking grid — typically red pegs for hits, white for misses — so you remember where you've already fired. The defender marks hits on their own ships too.

Hits and sinking

When all the cells of a single ship have been hit, that ship is **sunk**, and its owner must announce it ("You sank my battleship!"), usually naming which ship went down. That information matters tactically — knowing a 3-cell ship is fully sunk tells you the remaining hits belong to other ships.

In the classic rules each turn is exactly one shot regardless of whether it hits.

Notes

- Multiple games can be played at once, players can only match with one other player at a time.
- Players cannot play more than one game at a time
- Players should always have up to date information on who is online
- Players should always have up to date information on the leaderboard status
- If a player quits a match (logs out or closes the browser tab) then they automatically forfeit the match after a grace timer
- Grace timer for all timeout sequences (disconnected, waiting to accept game challenge) is 60 seconds
- each game has a unique id; the record holds both boards, the turn pointer, status, and an append-only move log of (player, x, y, result); socket identity comes from the login-token binding, never the payload; and reconnection replays the log filtered to that player's visibility.

Terms

User: a website user, has a username and password

Player: an in game user, the player has a map status that is kept track of by the game, players can be people or AI. the only difference is how the decisions are made.

Online: users are online if they logged in (this is going to be terrible later but whatevs)

Grace timer: time allowed for players to be disconnected from websockets before game is forfeited or challenge is abandoned

Game Terms

Hit: a single location of a ship is hit (triggers destruction animation 5 times)

Sunk: all locations of a ship have been hit (triggers appearance of ship on hidden state map for enemy player in all red, and marks ship on owner player as all red)